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UBC THE UNIVERSITY OF BRITISH COLUMBIA

Mapping BC's Solar Farm Potential: Executive Summary

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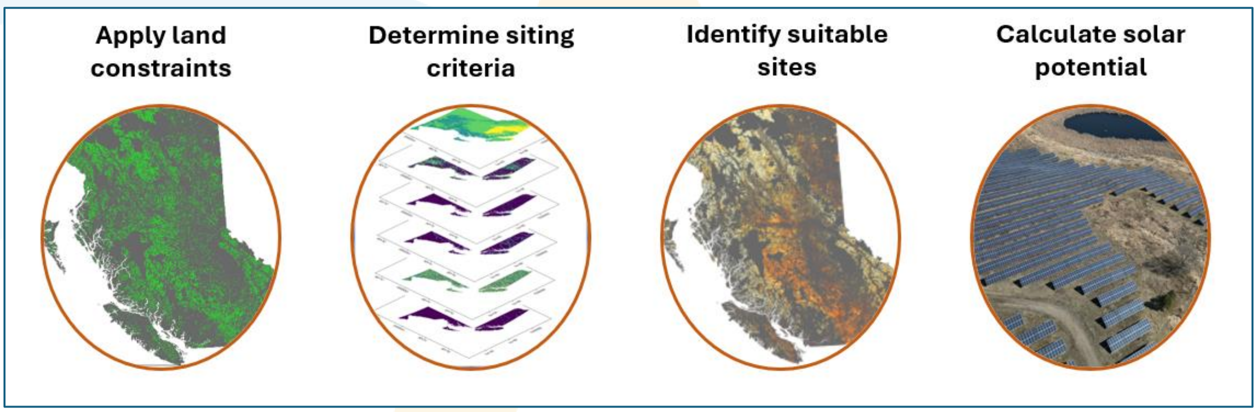
Purpose

To meet British Columbia's growing electricity demand over the next two decades while meeting climate commitments, the province must increase renewable energy infrastructure development. In response to this call to action, a GIS-based Analytic Hierarchy Process (AHP) analysis was performed to:

1. Identify locations across BC with the best potential for photovoltaic (PV) generation, and
2. Determine the total capacity and yearly energy generation potential possible from the suitable sites identified.

The model includes environmental, economic, and climatic criteria, together capturing core influences on renewable energy performance, accessibility, and development costs. As a screening tool, the results can guide organizations in prioritizing areas where development is most likely to yield high energy output.

Methods



The land constraint mask was created by identifying areas restricted from solar PV development for technical, environmental, or regulatory reasons. Individual constraint layers (Table 1) were converted into binary format and merged into a master exclusion zone, ensuring restricted lands were excluded from subsequent suitability assessments.

Table 1. The input layers that compromise the land constraints spatial mask.

Land Constraint	Portion of BC (%)
Unsuitable Land Covers – Water, Wetlands, Permanent Snow and Ice, Artificial/Built Environments	13.2
Dense Forests (Canopy Cover > 40%)	51.3
Protected and Conserved Areas	32.4
Linear Infrastructure (Roads/Railways)	2.4
Agricultural Land Reserve (ALR) with Soil Capability Class 1-3	0.84

To prepare a solar site suitability map from 5 (most suitable) to 1 (least suitable) six key criteria were considered (Table 1). These factors can be combined into a single suitability surface using the Analytical Hierarchy Process (AHP). To ensure model robustness, nine domain experts spanning academia, policy, and industry contributed to the weighting process through the individual creation of pairwise comparison matrices (PCM). Aggregating these PCMs via the geometric mean method grounded the weights in real-world planning priorities.

Table 2. The input criteria used to assess solar PV suitability across BC.

Criteria	Sub-Criteria	AHP-Derived Weight (%)
Climatic	Global Tilted Irradiance	26.3
Economic	Proximity to Substations	30.2
	Proximity to Demand Centers	14.4
	Proximity to Major Roads	11.2
Topographic	Ground Aspect	9.3
	Ground Slope	8.5

With the most optimal sites for PV development identified, the technical potential was determined for the top two suitability classes by calculating installable capacity (GW) and annual energy yield (TWh/year) for each viable site, using five parameters:

Parameter	Impact	Value
Site area (m^2)	Available space for arrays	Site dependent
Panel efficiency (η)	A measure how well a solar panel turns sunlight into usable electricity	23%
Latitude dependent Ground Cover Ratios (GCR)	Accounts for panel spacing and losses due to inter-row self-shading	Northern latitudes – 0.227 Central latitudes – 0.263 South latitudes – 0.314
Packing Density Ratio (R_{dt})	The ratio between direct and total land use in a designated site	0.724 (72%)
PV Power Potential (kWh/kW_p/year)	Long term yearly averages for standard crystalline silicon modules (c-Si) at a fixed optimum tilt (OPTA)	Site dependent

Technical yield results were generated for small scale solar farms (1 MW minimum production threshold) and large scale (100 MW minimum production threshold) by filtering out sites smaller than 5 acres and 500 acres, respectively. This follows the assumption that 5 acres of land is needed to generate 1 MW of solar energy.

Results

It was found that 75.5% of BC (716,261.8 km^2) is restricted from development due to unsuitable land cover types (e.g. water, built environments, wetlands), prime agricultural land, dense/mature forest stands, and protected and conserved areas. The remaining 24.5% of the province was classified as followed:

Table 3. Summary of the suitability distribution across the province

Suitability Class	Area (km^2)	Portion of Total Land Base (%)
1 – Least Suitable	80,926	8.53
2 – Less Suitable	108,095	11.40
3 – Moderately Suitable	39,216	4.13
4 – More Suitable	3,602	0.38
5 – Most Suitable	141	0.015

Resource yield estimates were summed across only the top two suitability classes (Class 4 and 5) as these represent the area with the most promising return on investment for solar PV development. Yield estimates (Table 3) represent the technical ceiling of energy generation and reflect a scenario where *all* sites deemed more and most suitable have installed solar arrays.

Table 4. Resource yield estimates across the top two suitability classes under two different minimum production threshold scenarios.

Scenario	1 MW	2 MW *Inside Municipal Boundaries	100 MW *Outside Municipal Boundaries
Number of sites	21,011	2755	93
Area of sites (km^2)	2843.5	668.0	433.9
Capacity (GW)	142.8	33.9	21.9
Total Energy (TWh)	182.3	42.0	28.9

Dissecting the results further, it was found that the Thompson-Nicola, East Kootenay, and Peace River regional districts hold the greatest amount of solar PV potential, as they achieved the biggest sum of Class 4 and 5 land, installable capacity, and annual energy generation out of the province.

Table 5. Summary of results for the top-ranking regional districts in terms of both geographical and technical potential. Results correspond to the baseline scenario with a 1MW production threshold.

Rank	Regional District	Area of Viable Sites (km^2)	Capacity (GW)	Annual Energy (TWh)
1	Thompson-Nicola	707.8	37.0	48.8
2	East Kootenay	460.8	24.1	32.0
3	Peace River	245.7	10.2	13.3

Conclusions

This project makes four key contributions to solar energy planning; first, it uses an expert-informed GIS-AHP framework, incorporating extensive engagement feedback from academics, industry professionals, and government/policy personnel into model parameters. Second, it advances beyond traditional suitability mapping by translating high-suitability areas into quantified annual energy generation estimates. Third, it offers an open-source decision-support tool which can be modified to fit future research needs (accessible through github). Lastly, it is the first solar PV suitability analysis for British Columbia - providing a tool to support the provinces transition to a diversified renewable energy portfolio.

Future Research

Future work should bridge spatial land-use suitability with electrical infrastructure constraints by considering utilization rate information. This would allow the model to prioritize substations that have greater availability than those that are at peak capacity. Another beneficial expansion would investigate and implement the effects of seasonal curtailment and battery storage options into the suitability model to further refine generation estimates. Ultimately, combining this GIS-based, land-use dominant evaluation, with dynamic grid modelling will transition the tool from regional site screen to an operation asset for energy developers.

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